Student ID_____ Student Name_____

IS221: Web Applications Development

Faculty of Science, Technology and Environment
School of Computing, Information and Mathematical Sciences

Final Examination-Semester I 2019

Face-To-Face and Online Mode

Duration of Exam: 3 hours + 10 minutes

Reading Time: 10 minutes

Writing Time: 3 hours

INSTRUCTIONS

1. This examination has four sections:
   a. Section A: Multiple Choice (20 marks)
   b. Section B: Coding Questions (34 marks)
   c. Section C: Short and Long Answer Questions (46 marks)
2. All questions are compulsory
3. Write ALL answers in the spaces provided within this paper.
4. This exam is marked out of 100 and is worth 50% of your overall course marks.
5. To pass this paper you require at least 40% for this exam which is equivalent to 40 marks out of 100
6. This paper has 25 pages including this page.

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Section A: Multiple Choice 20 Marks

Instructions:
Identify and circle the choice that best completes the statement or answers the question in the provided answer sheet on page 7. There is only **one** possible answer.

1) <body> is the ____ for the elements of a Web page.

   a. content box
   b. containing box
   c. containing block
   d. container

2) If the margin property has 4 values, the ____ margin is set to the first value, the ____ margin is set to the second value, the ____ margin is set to the third value, and the ____ margin is set to the fourth value?

   a. top; right; left; bottom
   b. right; left; top; bottom
   c. top; bottom; right; left
   d. top; right; bottom; left

3) If the value of the form method identified in the HTML Form element is Get, then the form results are HTML encoded and appended with a question mark to the URL requested as a single string called the ____?

   a. URL
   b. QueryString
   c. Form Fields
   d. Server Variables

4) The ____ control is used as a container to store dynamically added server controls.

   a. Placeholder
   b. Literal
   c. Page
   d. Panel

5) The ____ type input object clears the form of any user-entered data and returns it to its original state.
   a. redo
   b. rewrite
   c. reset
   d. clear

6) ____ let you move either to the previous or next page in the collection.

   a. Back and forth links
   b. Page turners
   c. Page changers
   d. Page changers

7) The inclusion of alt attributes is of prime importance to the ____ of your Web site.

   a. readability
   b. accessibility
   c. look and feel
   d. color scheme

8) ____ is simply a text document that contains the style rules.

   a. An internal style sheet
   b. The style attribute
   c. The <style> element
   d. An external style sheet

9) When you are designing a style sheet for a Web page, you can use ____ measurement values that adapt to the user's display type.

   a. fixed
   b. Absolute
   c. relative
   d. variable

10) Which of the following has the most weight in the style sheet order of precedence?

a. Author's style sheet
b. W3C's standard style sheet
c. User's style sheet
d. Browser's default style sheet

11) The ____ property gives you control over the specific sizing of your type.

a. font-style
b. font-size
c. font-family
d. font-variant

12) You can set ____ margin values to achieve special effects

a. zero
b. positive
c. negative
d. individual

13) If the paragraph element's margin-top value is 15px and margin-bottom value is 25px, the browser sets the vertical margin between paragraphs to ____ pixels.

a. 40
b. 25
c. 15
d. 10

14) You can disable the default underlining of hypertext links by setting the CSS text-decoration property to a value of ____.

a. no-underline
b. no-default
c. disable
d. none

15) An image's default hypertext border appears ____ before you click the image. The simplest type of selector is the ____ selector.

a. blue
b. red
c. black
d. purple

16) The simplest type of selector is the ____ selector.

a. type
b. grouping
c. combining
d. descendant

17) ____ positioning is where the element is entirely removed from the normal flow.

a. Relative
b. Absolute
c. Withdrawn
d. Removed

18) The ____ type input object sends the form data to the server using the transmission method specified in the <form> element.

a. text
b. reset
c. button
d. submit

19) Leading or trailing spaces cause problems when you try to join the contents of ____ cells.

a. leading
b. trailing
c. adjacent
d. all of the table

20) The following style rule selects an <a> element:

a. `{a font-size: 85% ; font-weight: bold; font-family: arial,sans-serif;}`
b. `a font-size: 85%; font-weight: bold; font-family: arial, sans-serif;`
c. `a {font-size: 85%; font-weight: bold; font-family: arial, sans-serif;}`
d. `a {font-size: 85%  font-weight: bold  font-family: arial, sans-serif}`

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Section A: Answer sheet 20 Marks

Circle the correct answer.

1.	**a**	**b**	**c**	**d**
2.	**a**	**b**	**c**	**d**
3.	**a**	**b**	**c**	**d**
4.	**a**	**b**	**c**	**d**
5.	**a**	**b**	**c**	**d**
6.	**a**	**b**	**c**	**d**
7.	**a**	**b**	**c**	**d**
8.	**a**	**b**	**c**	**d**
9.	**a**	**b**	**c**	**d**
10.	**a**	**b**	**c**	**d**
11.	**a**	**b**	**c**	**d**
12.	**a**	**b**	**c**	**d**
13.	**a**	**b**	**c**	**d**
14.	**a**	**b**	**c**	**d**
15.	**a**	**b**	**c**	**d**
16.	**a**	**b**	**c**	**d**
17.	**a**	**b**	**c**	**d**
18.	**a**	**b**	**c**	**d**
19.	**a**	**b**	**c**	**d**
20.	**a**	**b**	**c**	**d**

Section B: Coding Questions 34 Marks

Question 1: Data Tables

a. The table below shows my favorite Albums. Write the appropriate XHTML codes that will render the table as shown in Figure 1 below. The first line, My Favorite Albums, describes the table.

My Favorite Albums

Artist	Title	Comment
Hendrix	Electric Ladyland	Revolutionary
The Beatles	Revolver	Their best work.

Figure 1

(5 marks)

b. Write a style rule that can be applied to the table in figure 1 by adding the following properties:
   - Caption : left aligned
   - All borders collapse.
   - Font family: verdana
   - Table border: solid 2px green
   - Table cell border: solid 1px black
   - Table cells padding: 5 pixels **(5 marks)**

Question 2: Web Forms (XHTML)

Web forms let you build interactive web pages that collect information from a user and process it on the web server. The Web page in figure 2 below shows an example of a typical XHTML form. Use figure 2 web form and later information to answer the questions that follow.

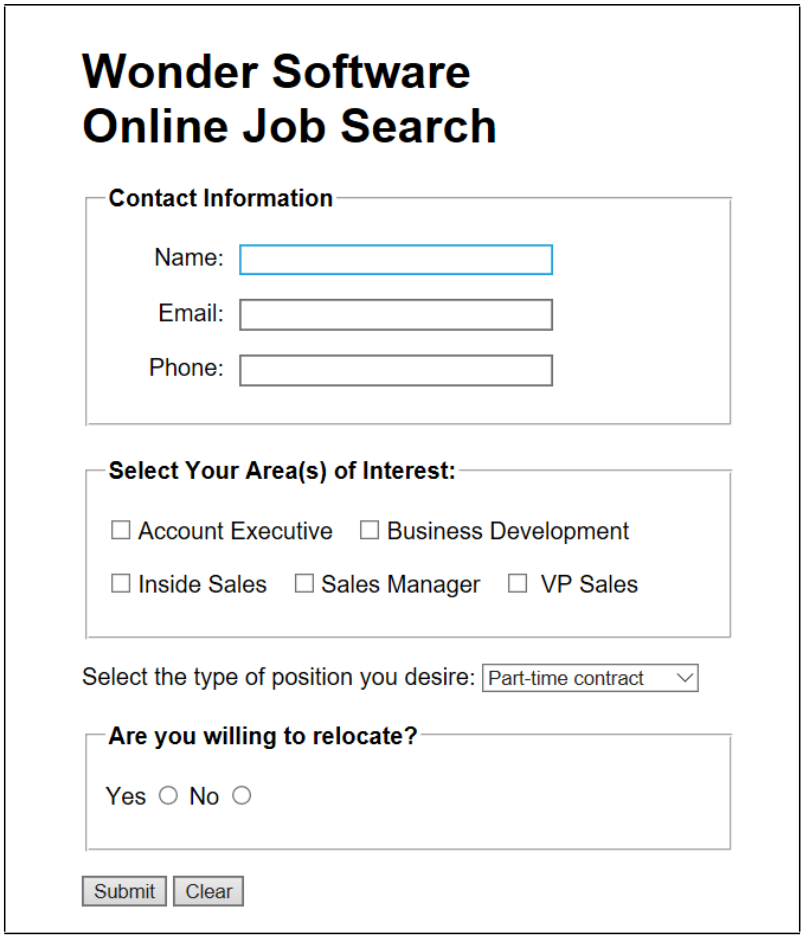

Figure 2

a. Identify the type of Input objects used in the form shown in figure 2. **(2 marks)**

b. Write the appropriate XHTML code(s) used to create the input objects: Submit and Clear as shown in figure 2. **(2 marks)**

c. The XHTML codes that create the Phone label and input object is given below. What is the purpose of adding the "for" attribute to the label element and "id" attribute to the input element. **(2 marks)**

```
<P><label for="phone" class="contact">Phone:</label>
    <input size="30" name="phone" id="phone"></p>
```

d. Write the XHTML codes that shows the VP Sales checkbox being checked in figure 2 form. **(2 marks)**

e. Write the complete XHTML codes of the last group "willing to relocate" in figure 2 form. The group is enclosed in a paragraph element. **(4 marks)**

Question 3: ASP.Net

a. The given code is for a button in ASP.NET.

```
<asp:Button ID="Button1" runat="server" Text="Button" />
```

The code for the above button will be rendered in the browser as XHTML codes. Write the XHTML codes that renders this button. **(2 marks)**

b. The code below often appears on top of each page or web form; it always defines page-specific attributes used by the ASP.NET page parser and compiler. Use the code below to answer the questions that follow

```
<%@ Page Language="VB" CodeFile=" Subscribe.aspx.vb" Inherits="
Subscribe" title=" Subscribe Page" AutoEventWireup="false" %>
```

i. What is the general name given to this code? **(1 mark)**

ii. What is the name of the web form? **(1 mark)**

iii. What is the name of the code behind page file? **(1 mark)**

c. Given below (in figure 3) is a web form named `ListControl.aspx` with two list controls. The web form also has a button, when clicked; it displays the selected items as text in a Label control.

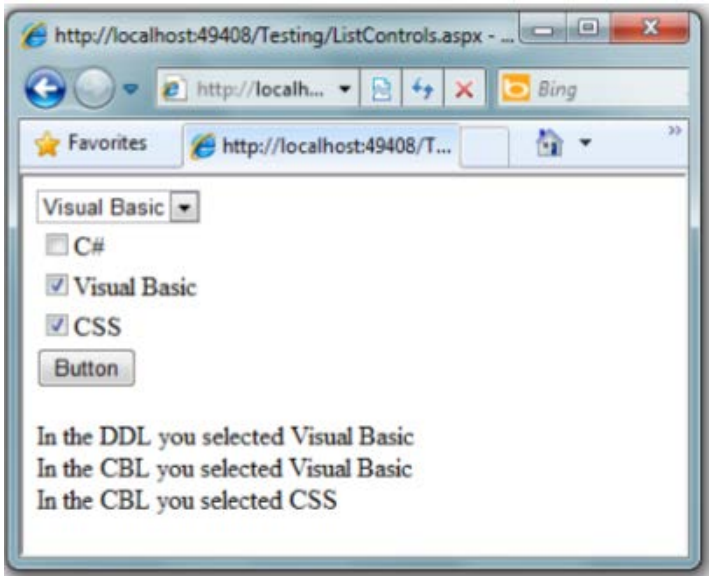

Figure 3

Incomplete code for the `ListControl.aspx` web form is given in Figure 4 below, with incomplete or missing sections indicated by **MISSING CODE**.

1	`<%@ Page Language="VB" AutoEventWireup="false" CodeFile="ListControls.aspx.vb" Inherits="ListControls" %>`
2	`<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">`
3	`<html xmlns="http://www.w3.org/1999/xhtml">`
4	`<head runat="server">`
5	`<title></title>`
6	`</head>`
7	`<body>`
8	`<form id="form1" runat="server">`
9	`<div>`
10	**`<!-- DropDownList Control -->`**

11	`<asp:DropDownList ID="DropDownList1" runat="server">`
	`<asp:ListItem>C#</asp:ListItem>`
	`<asp:ListItem>Visual Basic</asp:ListItem>`
	`<asp:ListItem>CSS</asp:ListItem>`
	`</asp:DropDownList>`

12	**<!-- CheckBoxList Control -->**
13	***MISSING CODE***
14	**<!-- Button Control -->**
15	<asp:Button ID="Button1" runat="server" Text="Button" />
16	
17	
18	**<!-- Label Control -->**
19	***MISSING CODE***
20	
21	</form>
22	</body>
23	</html>

Figure 4: ListControl.aspx web form source code

Use the ListControl.aspx web form source code in figure 4 above to answer the questions that follow

i. Write the missing codes at line 13 under the CheckBoxList control comment that renders the three check boxes as shown in figure 3. **(5 marks)**

ii. Write the missing codes at line 19 under the Label control comment that renders the comments below the button as shown in figure 3. **(2 marks)**

Section C: Short/Long Answer Questions 46 Marks

Question 1

What is a web application? **(2 marks)**

Question 2

What is the difference between static and dynamic web page? **(2 marks)**

Question 3

With the aid diagrams, discuss the differences between two -tier and three -tier Client/Server Architecture. Your discussion should include the following: **(6 marks)**

- Elements of two-tier system and their roles/functions
- How information flows in two-tier architecture
- Elements of three-tier system and their roles/functions
- How information flows in three-tier architecture

Question 4

What are the three methods to combine CSS style rules with XHTML code? Give an example of each method. **(6 marks)**

Question 5

Name the three space areas labelled as numbers 1, 2, and 3 as shown in figure 5 below.

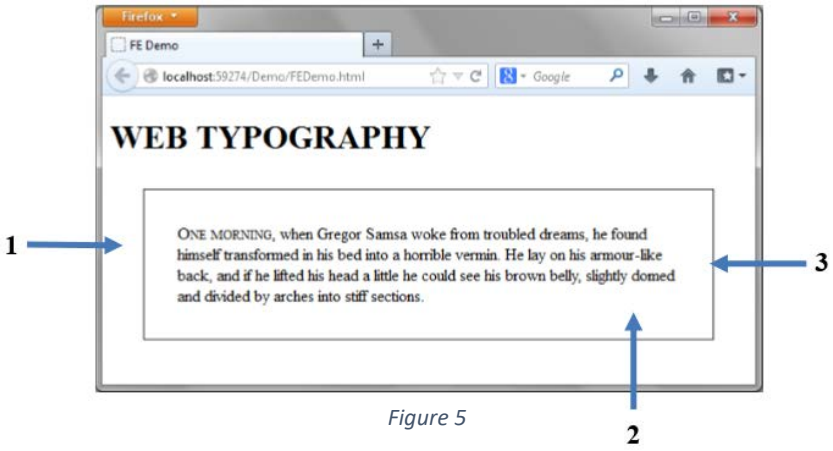

Figure 5

(3 marks)

Question 6

Write the appropriate CSS style that will be applied to Penguin image in figure 6 that will result in figure 7 on the right?

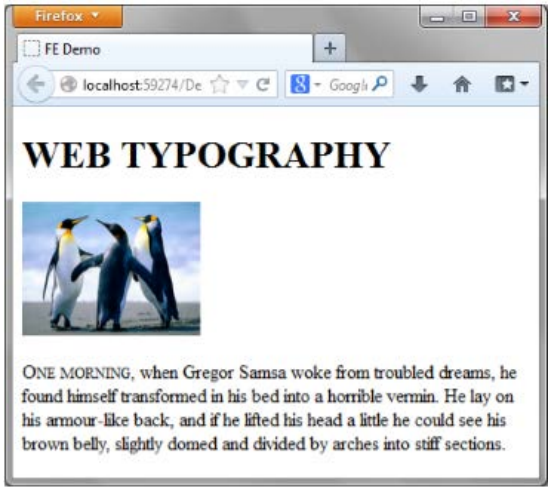

Figure 6

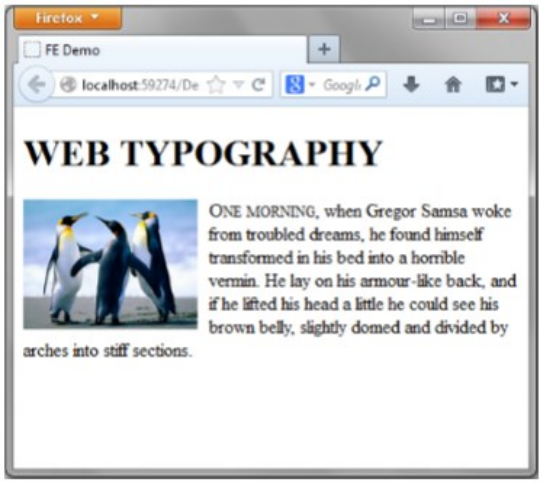

Figure 7

(2 marks)

Question 7

List four important guidelines when using images or graphics on your web page. **(2 marks)**

Question 8

You have learned of a number of web site design principles in this course. One important principle of web site design is to plan smooth transitions. Explain three ways to create smooth transition between pages of a web site. **(3 marks)**

Question 9

Effective navigation includes providing not only links to other pages in the Web site, but also cues to user's location. List the four navigation questions that users should be able to answer when navigating through your Web site. For each of the questions:

- Identify the type of information that the website is required to provide
- Identify tools/techniques that a website could use to provide this information

(8 marks)

Question 1:_____

Type of information:

_____

_____

Tool/technique:

_____

_____

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Question 2: _____

Type of information:

_____

_____

Tool/technique:

_____

_____

Question 3: _____

Type of information:

_____

_____

Tool/technique:

_____

_____

Question 4: _____

Type of information:

_____

_____

Tool/technique:

_____

_____

Question 10

Responsive design is a design strategy that is centered on designing your content so that it responds to the environment it is encountered and rendered in. Briefly discuss five techniques for responsive design that adapts to different screen sizes.

(5 marks)

Question 11

With the aid of a diagram, illustrate how you would design a responsive version of the homepage shown in figure 8 below for mobile devices that have a screen size of 320px.

(5 marks)

Figure 8

Question 12

Explain how the master and content pages are programmatically linked together? **(2 marks)**

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Web form Draft Page

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Draft Page

THE END